INCH-POUND
MIL-PRF-81757/9D
w/AMENDMENT 2
5 August 2021
SUPERSEDING
MIL-PRF-81757/9D
w/AMENDMENT 1
26 October 2017

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, 24 VOLTS, 30 AMPERE-HOUR (BB-433A/A)

Inactive for new design after 30 July 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery shall be M81757/9-2 or M81757/9-3. The M81757/9-2 is not for Navy use. The M81757/9-3 is a non-repairable, vented battery.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figure 1 of MIL-PRF-81757/9. The weight of the battery shall be not greater than 80.0 pounds (36.29 kg). Unless otherwise stated, all dimensions are in inches.

3. Capacity. The minimum capacity of the battery shall be not less than 30.0 ampere-hours (Ah). See 3.10.14 as modified herein.

4. Interface requirements. The battery shall contain 19 cells. The cells and other interior components of the battery shall be arranged and positioned as shown in either layout of figure 2 of MIL-PRF-81757/9.

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5. The receptacle terminal block shall be 0.866 ±0.015 from the centerline of the contact pins in the top surface of the terminal block. The receptacle shall be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MS3509-28 in accordance with MIL-PRF-18148/3.

6. All dimensions are prior to epoxy coating except: overall length = 11.812 max, overall width = 10.500 max., and overall height = 10.312 max.

7. Spacers, if used, shall be polyethylene sheet Classification PA0111 or PA0191 per ASTM D4066, minimum 0.015 inch thick, color neutral.

8. A sensor (thermocouple or thermistor), when required shall be located as shown on figure 2 for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and free side to side).

9. Length and width of cover pad and spacer shall conform to the inside dimensions of the applicable cover.

10. The length, width, and thickness of cover pad projections shall provide sufficient pressure on the vented filler caps to hold down the battery cells during vibration.

11. The quantity and location of cover projection pads shall be determined by the vented filler caps.

12. The base dimensions of the container shall be within tolerance for a distance of 2 inches up from the bottom.

13. The battery identification on the M81575/9-3 battery shall indicate the correct supersession date as indicated in Note 1.

14. The M81757/9-3 battery shall incorporate M81757/1-10 bolts to secure the cell interconnecting strap, thereby preventing cell removal.

15. The "CAUTION" label on the M81757/9-3 battery shall contain the following additional information: "The cells in this battery are not to be removed."

16. The cell vented filler cap shall be in accordance with MIL-PRF-81757/2.

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17. MIL-PRF-81757 compliance. The battery shall comply with MIL-PRF-81757 except as follows.

17.1 The following paragraphs of MIL-PRF-81757 are not applicable:

3.7.5 Handles.
3.10.10 Handle Strength.
3.10.20 Twenty-second pulse discharge.
3.10.22 Constant voltage discharge.
3.10.29 Storage effects.
4.5.9 Handle strength test.
4.5.20 Twenty-second pulse discharge test.
4.5.22 Constant voltage discharge.
4.5.29 Storage effects test.

18. Modify the following paragraphs:

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery container and container cover shall be as shown on figure 1 herein. The weight of the battery shall be not greater than 80 pounds (36.29 kg) after the inspection of 4.5.8 of MIL-PRF-81757."

3.10.14 Capacity. Delete the first sentence and insert: "The minimum capacity (see 6.9, 6.13.1, 6.13.2, and 6.13.14, all of MIL-PRF-81757) of the battery shall be not less than 30 Ah. C-rate discharges shall be conducted at 30 amperes."

3.10.14.1 Initial capacity. Delete the entire text and substitute: “To meet the initial capacity requirement, each cell shall produce a potential of not less than 0.95 volt at not less than 60 minutes into the constant 30-ampere discharge after being tested in accordance with 4.5.13 of MIL-PRF-81757 at room temperature (see 4.4.1 of MIL-PRF-81757).”

19. Modify the following paragraphs, tables, and figures of MIL-PRF-81757:

4.5.23 Cycling test. Delete the test of 4.5.23.b(1) and insert: “(1) Discharge the battery for 20 seconds at constant current load of 600 amperes.

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TABLE III. Qualification inspection of batteries. Delete table III of MIL-PRF-81757 and insert the table shown below.

TABLE III. Qualification inspection of batteries.

Test number	Examinations and tests	Sample number					Requirement paragraph 1/	Test paragraph 1/
		1	2	3	4	5		
1	INSPECTION FOR DAMAGE	X	X	X	X		3.10.1 (G)	4.5.1 (G)
2	VISUAL AND MECHANICAL EXAM	X	X	X	X		3.10.6 (G), reqmt 4 (N) & 5 (N)	4.5.6 (G)
3	DIMENSIONS AND WEIGHT	X		X			3.10.8 (S), reqmt 2 (N)	4.5.8 (G)
4	CELL VENT TEST	X	X	X	X		3.10.13 (G)	4.5.12 (G)
5	INITIAL CAPACITY DISCHARGE	X	X	X	X		3.10.14 (S), reqmt 3 (N)	4.5.13 (G)
6	HUMIDITY AND CHARGE RETENTION TEST			X	X		3.9 (G), 3.10.15 (G)	4.5.14 (S)
7	TEMPERATURE SHOCK TEST			X	X		3.9 (G), 3.10.17 (G)	4.5.17 (G)
8	ALTITUDE TEST			X	X		3.9 (G), 3.10.18 (G)	4.5.18 (G)
9	SALT FOG TEST	X	X				3.9 (G), 3.10.19 (G)	4.5.19 (G)
10	CYCLING TEST	X	X				3.9 (G), 3.10.23 (S)	4.5.23 (S)
11	TEMPERATURE RISE AND FLOAT TEST	X	X				3.9 (G), 3.10.9 (G), 3.10.24 (G)	4.5.24 (S)
12	PHYSICAL INTEGRITY AT HIGH TEMPERATURE	X					3.9 (G), 3.10.25 (G)	4.5.25 (G)
13	INTERNAL PRESSURE TEST	X					3.9 (G), 3.10.26 (G)	4.5.26 (G)
14	SHOCK TEST (BASIC DESIGN)			X	X		3.9 (G), 3.10.16 (G)	4.5.15 (G)
15	VIBRATION TEST			X	X		3.9 (G)	4.5.16 (G)
16	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	X		3.10.27 (G)	4.5.27 (G)
17	CHARGE STABILITY TEST (REQUIRED FOR BATTERIES WITH GAS BARRIER AND SEPARATOR SYSTEM MATERIALS NOT LISTED IN 6.10.4 (G))		X	X			3.9 (G), 3.7.11.4 (G), 3.10.24 (G), 3.10.28 (G)	4.5.28 (G)
18	STORAGE EFFECTS TEST					X	3.9 (G), 3.10.1 (G), 3.10.9 (G), 3.10.6 (G), 3.10.14 (S), 3.10.24 (G), 3.10.29 (G), reqmt 3 (N)	4.5.29 (G)
19	SHELF LIFE TEST	X			X		3.9 (G), 3.10.14 (S), 3.10.30 (G), reqmt 3 (N)	4.5.30 (G)
20	CELL BAFFLE TEST	SAMPLE CELL OF 4.2.1 (G)					3.7.11.5 (G), 3.10.12 (G)	4.5.11 (G)
21	TAMPER-RESISTANT FASTENER TEST FOR FASTENERS OTHER THAN SHEAR-HEAD SCREWS	SAMPLE CELL OF 4.2.1 (G)					3.7.9 (G)	4.5.31 (G)

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1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph is as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by MIL-PRF-81757/9. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-81757/9.

TABLE IV. Qualification inspection of components and materials. Delete and substitute the following;

TABLE IV. Qualification inspection of components and materials.

Part	Examinations and tests	Requirement paragraph 1/	Test paragraph 1/
Metal parts 2/	Electrolyte resistance	3.8.1 (G), 3.8.4 (G), 3.9 (G), 3.10.2 (G)	4.5.2 (G)
Components 3/, elastomeric materials	Electrolyte resistance; dielectric strength and insulation resistance	3.8.1 (G), 3.8.4 (G), 3.9 (G), 3.10.3 (S), 3.10.27 (G)	4.5.3 (G) 4.5.27 (G)
Plastic parts	Electrolyte absorption	3.8.5 (G), 3.9 (G), 3.10.4 (G)	4.5.4 (G)
Vented filler caps	Electrolyte resistance; vented filler cap test	3.9 (G), 3.10.3 (G) 3.10.5 (G), 3.10.6 (G)	4.5.3 (G), 4.5.5 (G), 4.5.6 (G)

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph is as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by MIL-PRF-81757/9.

2/ Including but not limited to the container, cover, and latches; external cell terminal hardware; and intercell connectors and hardware.

3/ Including but not limited to the vented filler caps, receptacles, and parts made of elastomers.

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TABLE VII. Group B inspection. Delete table VII of MIL-PRF-81757 and substitute the table shown below.

TABLE VII. Group B inspection.

Test no.	Examinations and tests 1/	Requirement paragraph 1/	Test paragraph 1/
	Subgroup I:		
I-1	Dimensions and weight	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
I-2	Vented filler cap test (two vented filler caps per battery)	3.6(G), 3.7.12(G), 3.8.6(G), 3.9(G), 3.10.5 (G)	4.5.5 (G)
	Subgroup II:		
II-1	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
	Subgroup III:		
III-1	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
III-2	Internal pressure test	3.9 (G), 3.10.26 (G)	4.5.26 (G)

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph is as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by MIL-PRF-81757/9. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-81757/9.

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TABLE VIII. Group C inspection. Delete table VIII of MIL-PRF-81757 and insert the table below.

TABLE VIII. Group C inspection.

Test no.	Examinations and tests	Requirement paragraph 1/	Test paragraph 1/
I-1	Subgroup I: Inspection for damage	3.10.1 (G)	4.5.1 (G)
I-2	Visual and mechanical exam	Reqmt 4 (N) and 5 (N); 3.10.6 (G)	4.5.6 (G)
I-3	Dimensions and weight (weigh only after adjustment)	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
I-4	Cell vent test	3.10.13 (G)	4.5.12 (G)
I-5	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
I-6	Cycling test	3.9 (G), 3.10.23 (S)	4.5.23 (G)
I-7	Temperature rise and float test	3.9 (G), 3.10.9 (G), 3.10.24 (G)	4.5.24 (G)
II-1	Subgroup II: Inspection for damage	3.10.1 (G)	4.5.1 (G)
II-2	Visual and mechanical exam	Reqmt 4 (N) and 5 (N); 3.10.6 (G)	4.5.6 (G)
II-3	Dimensions and weight (weigh only after adjustment)	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
II-4	Cell vent test	3.10.13 (G)	4.5.12 (G)
II-5	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
III-1	Subgroup III: Inspection for damage	3.10.1 (G)	4.5.1 (G)
III-2	Visual and mechanical exam	Reqmt 4 (N) and 5 (N); 3.10.6 (G)	4.5.6 (G)
III-3	Dimensions and weight (weigh only after adjustment)	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
III-4	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
III-5	Altitude	3.9 (G), 3.10.18 (G)	4.5.18 (G)
III-6	Medium rate discharge, operating position	3.9 (G), 3.10.21 (G)	4.5.21 (G)

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph is as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by MIL-PRF-81757/9. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-81757/9.

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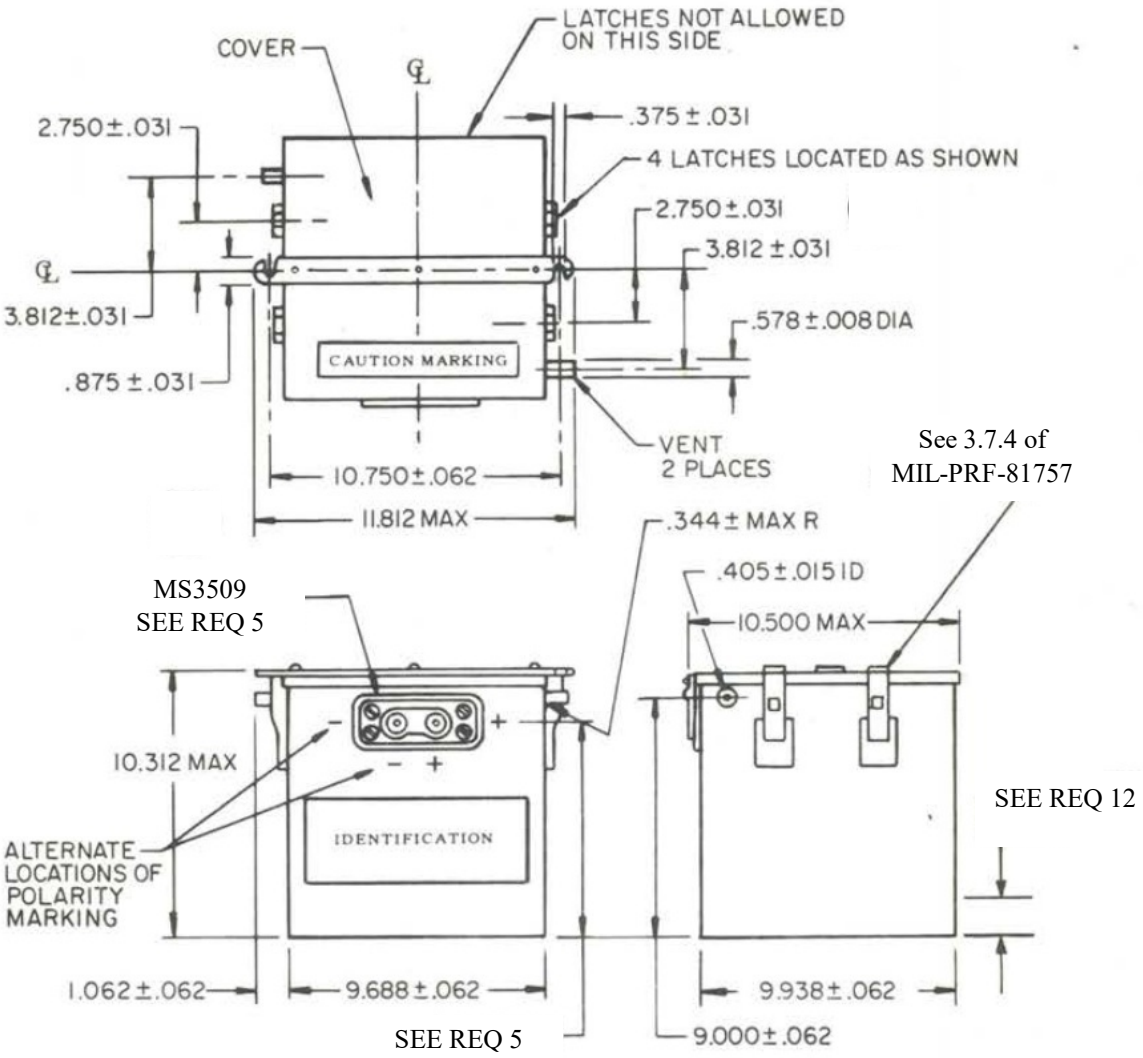


FIGURE 1. Dimensions and configuration for PIN M81757/9-2 and M81757/9-3.

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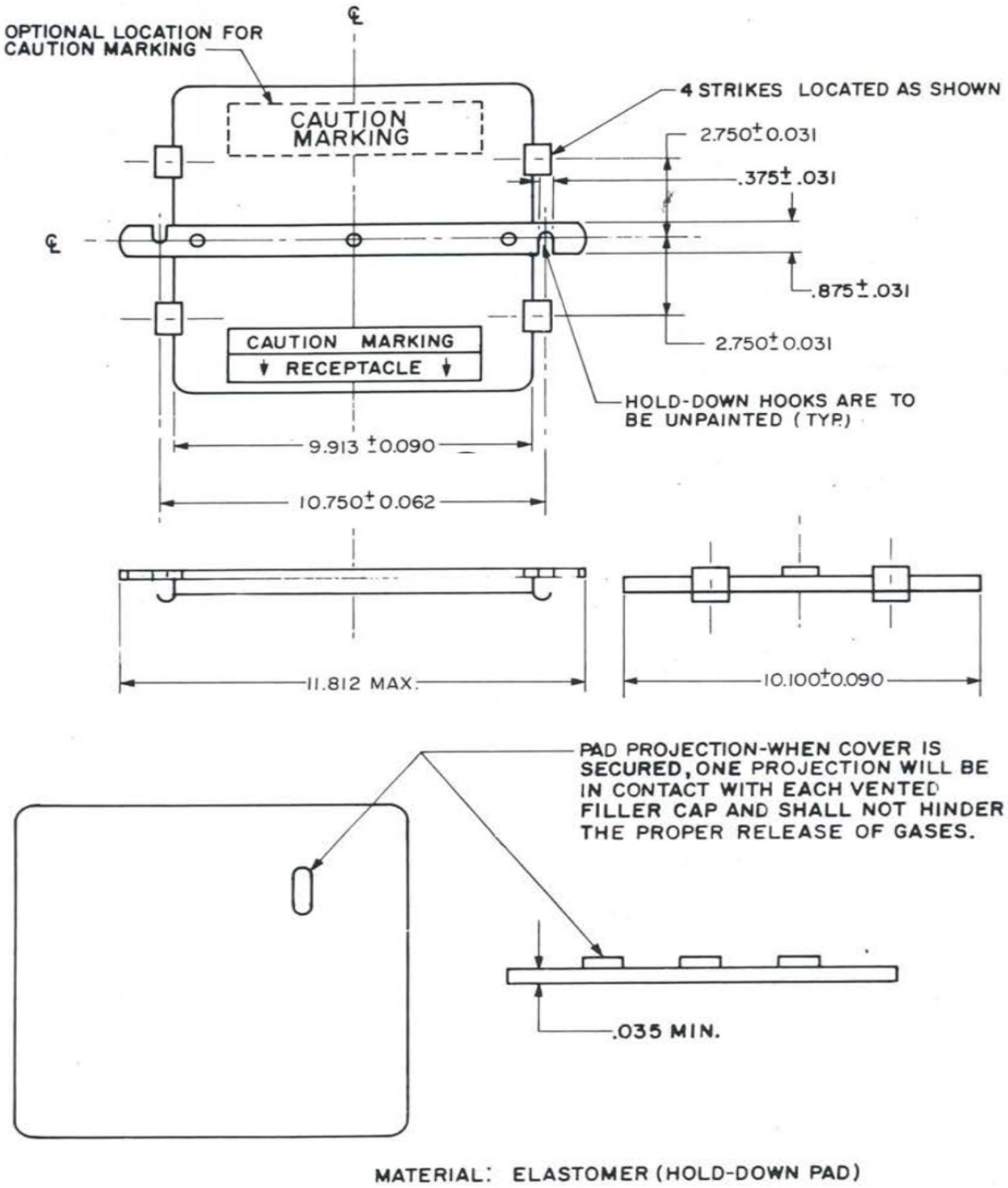


FIGURE 1. Dimensions and configuration for PIN M81757/9-2 and M81757/9-3 - Continued.

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CONNECTOR STRAP ARRANGEMENT

REFERENCE LETTER	STRAP PART NUMBER	DESCRIPTION	QUANTITY
A	M81757/3-5	Connector, Strap	12
B	M81757/3-6	Connector, Strap	6
C	M81757/4-2	Connector, 2 Tab	2

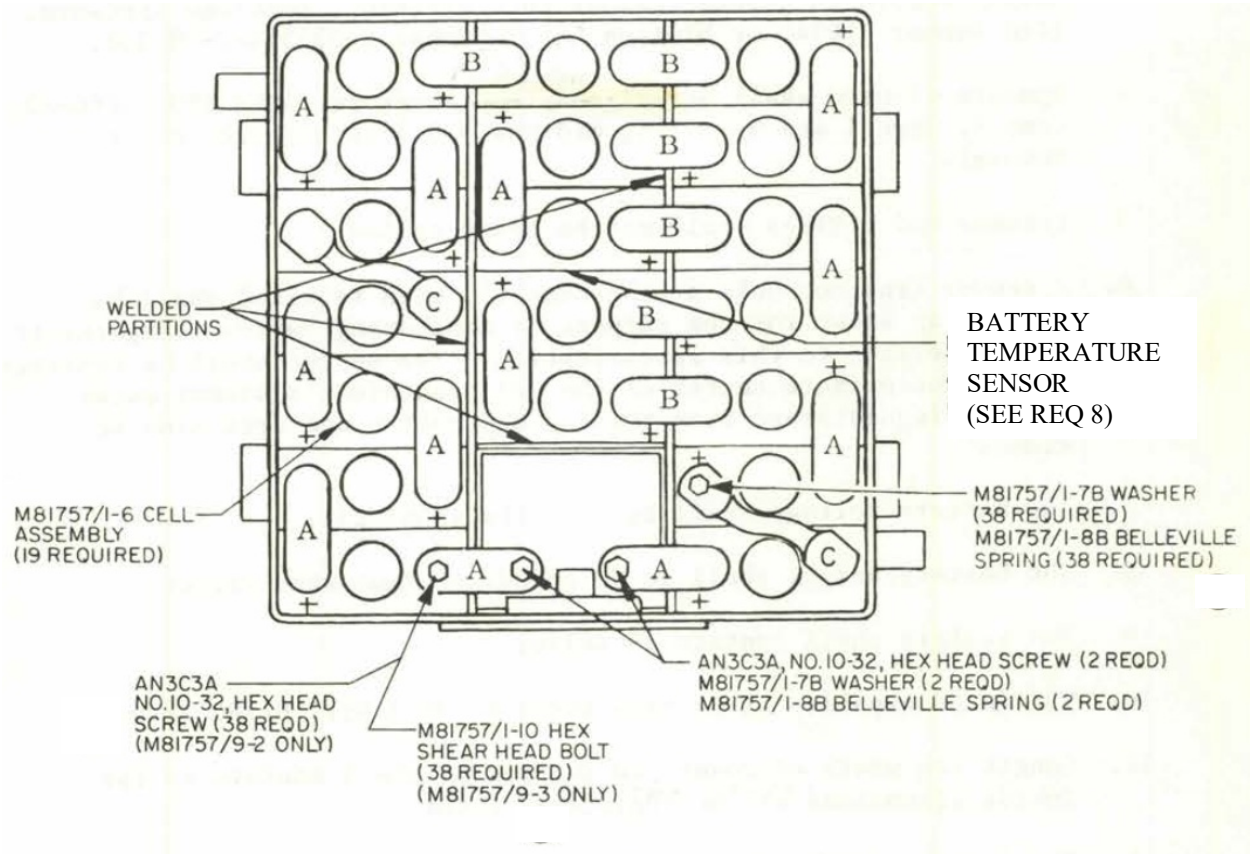


FIGURE 2. Layout (reference) of interior components for PIN M81757/9-2 and M81757/9-3.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT REPAIRABLE, NICKEL-CADMIUM
MILITARY PIN:	M81757/*** ***
TYPE DESIGNATION:	
NATIONAL STOCK NUMBER:	6140-**-***-****
REPLACES:	***
CELL REQUIRED:	*** EACH M81757/***, NSN ***
*** VOLTS	NOMINAL VOLTAGE:
MAXIMUM WEIGHT:	*** POUNDS
INITIAL CAPACITY:	*** AH (*** AMPS/***MINUTES/77 °F/*** V)
MINIMUM CAPACITY:	*** AH/1 HR/77 °F/*** V
CONTRACT AND DELIVERY ORDER NUMBER:	***
MANUFACTURED BY:	***
MANUFACTURER CAGE:	***
MANUFACTURER SERIAL NUMBER:	***
DATE/LOT CODE:	***

FIGURE 3. Marking for M81757/9-2 and M81757/9-3 batteries.

NOTES:

1. Supersession data: This specification sheet supersedes MS18045, MS90321, MS24498 and MIL-B-81757 part numbers as follows, when a product has been qualified for each MIL-PRF-81757 part number respectively.

Part Number	Replaces Part Numbers
M81757/9-2	MS24496-1, MS24498-2
M81757/9-3	MS18045-46, MS18045-50, M81757/9-2, MS90321-75W, MS90321-69W

Part Number	Superseded by
MS18045-46	M81757/9-3
MS18045-50	M81757/9-3
MS24498-1	M81757/9-2
MS24498-2	M81757/9-2
M81757/9-2	M81757/9-3
MS90321-75W, MS90321-69W	M81757/9-3

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Referenced documents. In addition to MIL-PRF-81757, this specification sheet references: ASTM D4066, MIL-PRF-18148/3, and MIL-PRF-81757/2.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - AS
Air Force - 20
DLA - CC

Preparing activity:
Navy - AS
Project 6140-2021-006

Agent:
Navy - SH

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